

associability_analysis

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.2      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```
library(broom.mixed)
```

Run associability_models.R first.

```
m1 <- readRDS("m1.RDS")
m2 <- readRDS("m2.RDS")
m3 <- readRDS("m3.RDS")
```

```
anova(m1, m2, m3)
```

```
## Data: data
## Models:
## m1: old_chosen ~ old_value_centered + true_deck_centered + (old_value_centered + true_deck_centered
## m2: old_chosen ~ old_value_centered + true_deck_centered + (old_value_centered + true_deck_centered
## m3: old_chosen ~ old_value_centered + true_deck_centered + (old_value_centered + true_deck_centered
##      npar    AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m1      9 58584 58663 -29283      58566
## m2     11 58582 58678 -29280      58560 6.4706  2    0.03935 *
## m3     12 58582 58687 -29279      58558 1.6395  1    0.20040
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

m2 is the preferred model. This includes object random effects, suggesting that there is a significant value associability effect. Here we can visualize how big this effect is by looking at the change from a low associability object image to a high associability object image.

```

coefs <- tidy(m2) |>
  filter_out(str_detect(group, "sub_factor")) |>
  select(term, estimate) |>
  deframe()

ggplot() +
  xlim(-.5, .5) +
  geom_function(aes(color = "Average Object",
    fun = \(x) plogis(coefs["(Intercept)"] + coefs["old_value_centered"] * x)) +
  geom_function(aes(color = "+2 SD Object",
    fun = \(x) plogis(coefs["(Intercept)"] +
      (coefs["old_value_centered"] * x) +
      (2 * coefs["sd__old_value_centered"] * x))) +
  geom_function(aes(color = "-2 SD Object",
    fun = \(x) plogis(coefs["(Intercept)"] +
      (coefs["old_value_centered"] * x) -
      (2 * coefs["sd__old_value_centered"] * x))) +
  annotate("text", x = .3, y = c(.68, .65, .62),
    label = c("+2 SD Object", "Average Object", "-2 SD Object")) +
  theme_classic() +
  theme(legend.position = "none")

```

